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SOC Analyst · CompTIA Security+ · Network+ (in progress) · Purdue Cybersecurity '26

Sec+

CompTIA Security+ certified

2+ yrs

IT & network experience

Meta

Enterprise IT support (via 2020 Companies)

15

Public security & code repos

235

Live metrics in Zabbix lab

01 · FIREWALL

pfSense Segmentation

WAN / LAN / DMZ design with DHCP reservations, BIND DNS zones, and interface firewall rules.

02 · MONITORING

PRTG Service Health

Ping, DNS v2, and HTTP sensors tracking latency, resolution time, and availability.

03 · INFRASTRUCTURE

Zabbix Living Lab

Enterprise monitoring of a live server-room hypervisor: 235 metrics, 89 triggers, email alerting.

pfSense Firewall, Segmentation, DHCP & DNS

pfSense 2.7.2

BIND DNS

DHCP

DMZ

VMware

NETWORK DESIGN



WHAT I BUILT

- Segmented VMs into **LAN** and **DMZ** zones with static interface addressing for traffic isolation
- Wrote interface-based firewall rules on the DMZ (OPT1) to control what can cross zones
- Ran DHCP on both zones (pool .100-.200) with **MAC-based static reservations** for DMZ servers (web 192.168.2.20, mail .30)
- Installed **BIND** and built forward/reverse zones with A, NS, MX, CNAME, and PTR records
- Set upstream forwarders (8.8.8.8, 1.1.1.1) and SMTP alert notifications
- Hardened** by disabling the unused DNS Forwarder/Resolver and DHCP on interfaces that didn't need them
- Troubleshooted webConfigurator access and recovered admin credentials from the console

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Starting CRON... done.
pfSense 2.7.2-RELEASE amd64 20240304-1953
Bootup complete

FreeBSD/amd64 (ashrall.ashrall.cit) (ttyv0)

VMware Virtual Machine - Netgate Device ID: a227911c7b

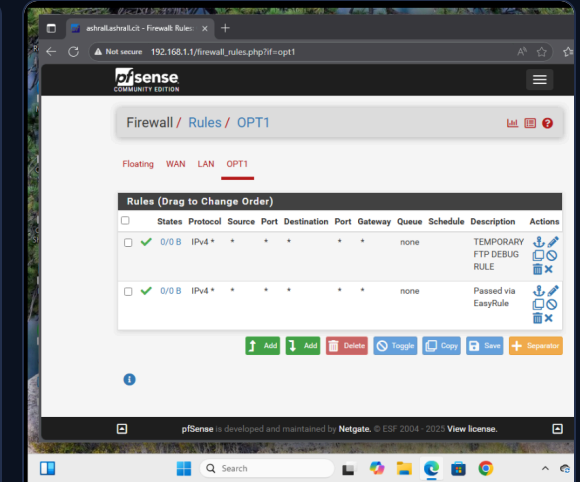
*** Welcome to pfSense 2.7.2-RELEASE (amd64) on ashrall

WAN (wan)      -> em0      -> v4/DHCP4: 192.168.11.132/24
LAN (lan)      -> em1      -> v4: 192.168.1.1/24
OPT1 (opt1)    -> em2      -> v4: 192.168.2.1/24

0) Logout (SSH only)          9) pfTop
1) Assign Interfaces          10) Filter Logs
2) Set interface(s) IP address 11) Restart webC
3) Reset webConfigurator password 12) PHP shell +
4) Reset to factory defaults  13) Update from
5) Reboot system              14) Enable Secur
6) Halt system                 15) Restore rece
7) Ping host                   16) Restart PHP-
8) Shell
  
```

Enter an option: █

Console · WAN / LAN / OPT1 assignment



Firewall > Rules > OPT1 (DMZ)

The screenshot shows the BIND Zone Domain records configuration. It lists several records for the 'pfsense' and 'pfsense-dmz' domains, including A records for 'pfsense', 'pfsense-dmz', 'web', and 'mail', and a CNAME record for 'www'. It also shows MX records for '@' and 'mail'.

BIND zone records · A / NS / MX / CNAME

The screenshot shows the DHCP specification configuration. It displays the 'Primary Address Pool' with a subnet of 192.168.1.0/24 and a subnet range of 192.168.1.1 - 192.168.1.254. The address pool is set to 192.168.1.100. There is an option to 'Add Address Pool'.

DHCP address pool · 192.168.1.100-200

PRTG Network Monitor: Service Health & Availability

PRTG Ping DNS v2 HTTP Log review

7 ms

Ping latency · 0% packet loss

92 ms

DNS v2 resolution time

482 ms

HTTP page load time

100%

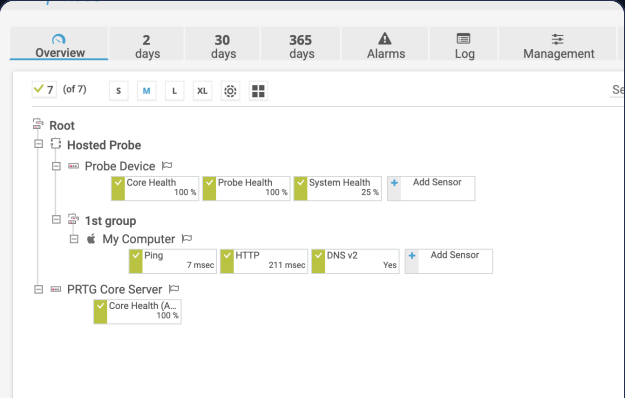
Uptime across all sensors

WHAT I DID

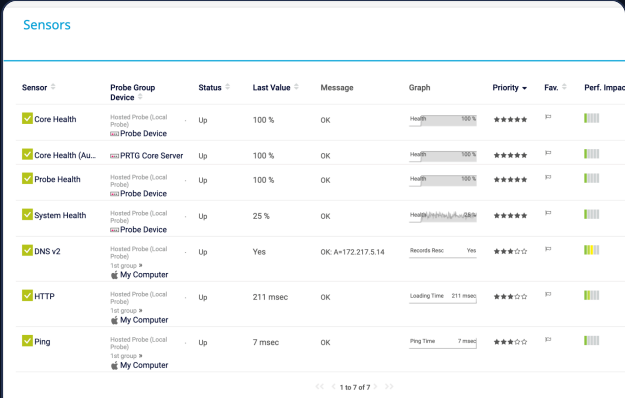
- ▶ Deployed **Ping**, **DNS v2**, and **HTTP** sensors, each covering a different failure mode: connectivity, name resolution, and application availability
- ▶ Organized the device tree (Hosted Probe → group → device) so health is visible at a glance
- ▶ Baselined performance: flat latency and no jitter or packet loss, giving a known-good reference for spotting anomalies
- ▶ Reviewed sensor logs (Created → Unknown → Up) to build a timestamped audit trail for troubleshooting
- ▶ Kept sensor performance impact low so monitoring doesn't load the host

WHY IT MATTERS FOR A SOC

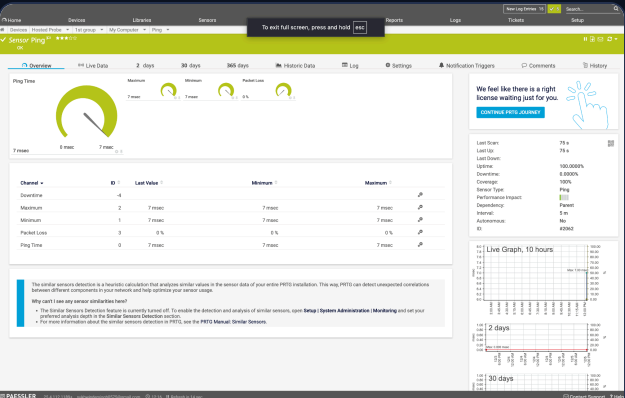
A security analyst needs a baseline to tell a problem from normal. These sensors define "normal" for a service, so an outage, a DNS failure, or a slow response stands out right away.



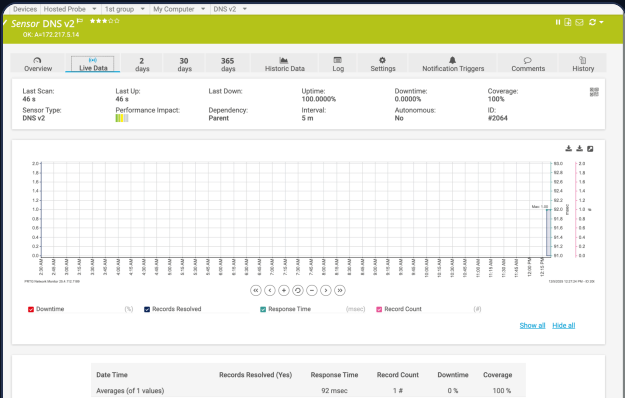
Device tree · all sensors Up



Sensor list · status & last value



Ping sensor · 7 ms, 0% loss



DNS v2 · 92 ms, 100% uptime

Zabbix Monitoring of a Live Server Room

Zabbix 7.0 SNMP v2c iptables XCP-ng 8.3 Kali Linux

235

Metrics monitored 24/7

89

Alert triggers configured

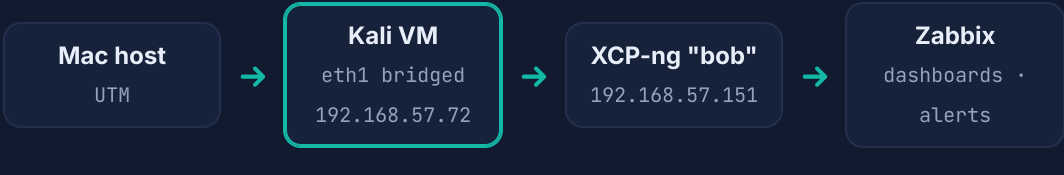
1.5 ms

Latency to server · 0% loss

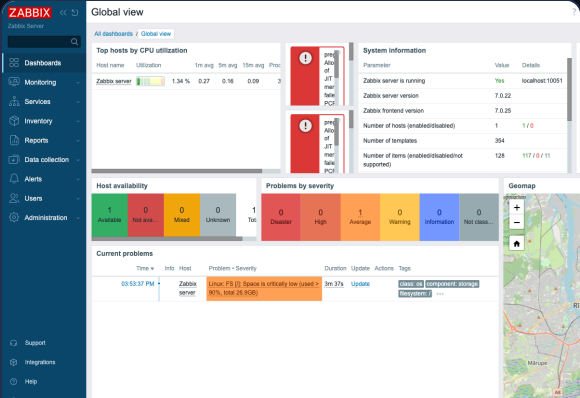
60 s

SNMP polling interval

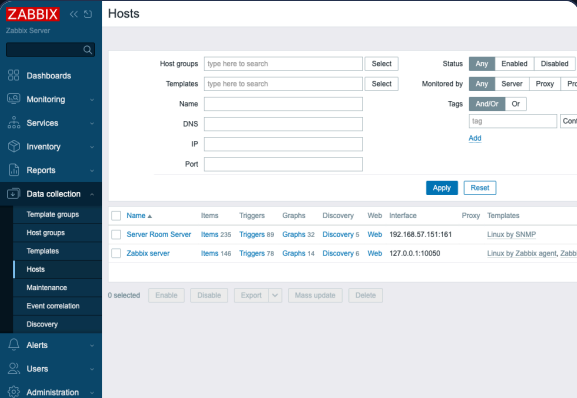
MY ROLE: INFRASTRUCTURE & SNMP LEAD



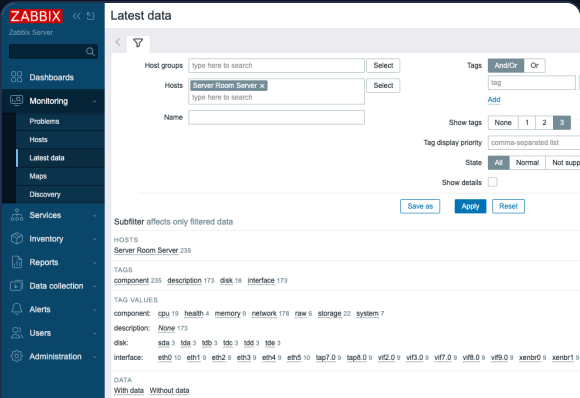
- ▶ **Solved the blocker:** the VM was stuck behind NAT (192.168.64.x). I added a bridged NIC on a USB Ethernet adapter to put it directly on the server-room subnet
- ▶ **Built the stack from scratch:** Zabbix 7.0 server and frontend on Apache2, MariaDB, and PHP 8, with every setup check passing
- ▶ **Configured SNMP on a production hypervisor:** wrote snmpd.conf and restricted access to 192.168.57.0/24
- ▶ **Least-privilege firewalling:** opened only UDP/161 in iptables and made the rule survive reboots
- ▶ **Verified end to end** with snmpwalk, with live CPU, memory, disk, and interface data in Zabbix plus automated email alerts



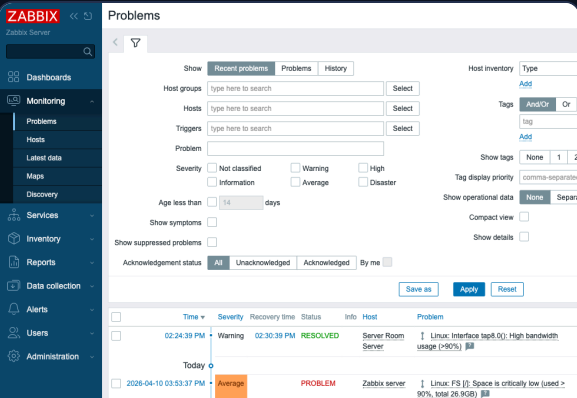
Global dashboard · host availability



Server Room Server · 235 items via SNMP



Latest data · live CPU / RAM



Problems · triggered & resolved alerts

Security Tooling & Labs

[Python](#) [Suricata](#) [Wireshark](#) [Nmap](#) [Cryptography](#)

DETECTION & DEFENSE

<code>firewall-log-analyzer</code>	Python CLI for SOC firewall-log triage: parser, timezone normalization, IPv4 classification, and geo/deny analytics with unit tests. 44K-event dataset .
<code>ddos-attack-detection-lab</code>	Simulated SYN/UDP floods and Slowloris, then detected them with a Suricata IDS and Wireshark analysis.
<code>linux-firewall-and-file-integrity</code>	Default-deny stateful iptables firewall plus a SHA-256 file-integrity / tamper-detection monitor.
<code>endpoint monitoring (Snort)</code>	Detected suspicious endpoint and network activity with Snort IDS , mapped simulated attacks to MITRE ATT&CK , and wrote IR procedures.
<code>healthnetwork-security-assessment</code>	12-asset threat model with CIA classification and Nmap reconnaissance from Kali.

AUTOMATION & CRYPTO

<code>breach-exposure-analyzer</code>	Triages workforce breach exposure, rates risk LOW / MEDIUM / HIGH, and auto-drafts notification emails .
<code>osint-recon-toolkit</code>	Passive recon: DNS resolution and attack-surface enumeration via Shodan InternetDB and scraping.
<code>rsa-key-rotation</code>	Bulk re-encryption from a retired RSA key to a new one (RSA-OAEP / SHA-256).
<code>phone-number-intelligence</code>	Validates numbers against NANP rules from SQLite and searches by number, area code, exchange, or name.
<code>sha1-collision-attack</code>	Reproduced the SHATTERED SHA-1 collision with a verification script and an IEEE-style paper.

Open to: SOC Analyst · Security Analyst · Network Security roles · Indianapolis / Hybrid / Remote

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